THE REDUCTION OF A COMPLETION OF A FORMAL ALGEBRAIC SPACE IS THE REDUCED CLOSED SUBSPACE IT IS TAKEN ALONG

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The reduction of a formal algebraic space and its universal property. Same chapter, Lemma lemma-reduction-formal-algebraic-space, invoked at proof lines 8080-8082. Gives the mapping property of the reduction of the completion.
- The reduced induced closed subspace of a closed subset and its universal property. Properties of Spaces, Definition spaces-properties-definition-reduced-induced-space and Lemma spaces-properties-lemma-reduced-closed-subspace, invoked at proof lines 8082-8086. Gives the mapping property of the reduced closed subspace.
- The definition of the formal completion of a formal algebraic space along a closed subset of its reduction. Same chapter, the definition of the completion functor, in force in the statement. Supplies the condition that the image of a morphism lies in the closed subset.

Source reference. Original lemma 0AIN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2833–2833. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 047X: sources/stacks-chow/tags.txt, source lines 5524–5524. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03IQ: sources/stacks-chow/tags.txt, source lines 4612–4612. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *Let S be a scheme. Let X be a formal algebraic space over S. Let $T \subset |X_{red}|$ be a closed subset. The reduction $(X_{/T})_{red}$ of the completion $X_{/T}$ of X along T is the reduced induced closed subspace Z of X_{red} corresponding to T.*

Proof. It follows from Lemma 0AIN, Properties of Spaces, Definition 047X (which uses Properties of Spaces, Lemma 03IQ to construct Z), and the definition of $X_{/T}$ that Z and $(X_{/T})_{red}$ are reduced algebraic spaces characterized the same mapping property: a morphism $g : Y \to X$ whose source is a reduced algebraic space factors through them if and only if $|Y|$ maps into $T \subset |X|$. □